Lecture 21 Piecewise Interpolation and Numerical Differentiation @April 19, 2026

Summary

Executive Summary

- Piecewise polynomial interpolation addresses the oscillation and non-locality problems inherent in high-order global polynomial interpolation by dividing the interval into subintervals and constructing separate polynomials on each.
- Cubic splines provide C^2 continuity at interior nodes (matching function value, first derivative, and second derivative) while using only local data, making them significantly smoother than piecewise linear interpolation.
- Natural cubic splines (second derivative zero at endpoints) and clamped cubic splines (first derivative specified at endpoints) both have unique existence guaranteed by linear algebra arguments on well-conditioned systems.
- The cubic spline minimizes total oscillation: $int_a^b (S''(x))^2 , dx leq int_a^b (f''(x))^2 , dx$, bounding spline oscillations by the original function's oscillations—a key advantage over global polynomial interpolation.
- Numerical differentiation via finite differences (e.g., forward difference $f'(x_0) approx frac{f(x_0+h)-f(x_0)}{h}$) faces a fundamental tradeoff: smaller h reduces truncation error ($O(h)$) but amplifies round-off error (proportional to $epsilon/h$), requiring optimal step size $h_{text{opt}} sim sqrt{epsilon}$.
- Taylor series and polynomial interpolants are the primary tools for deriving finite difference formulas and estimating truncation error; applications include discretizing ODEs and PDEs.

Conceptual Map (core themes)

(A) Core Concept & Context

Piecewise polynomial interpolation replaces a single high-order polynomial over $[a, b]$ with lower-order polynomials on subintervals $[x_j, x_{j+1}]$. This resolves

two flaws of global interpolation: (1) inability to isolate behavior in one part of the interval from another, and (2) Runge-like oscillations when polynomial degree increases.

(B) Detailed Explanation

Instead of constructing one polynomial through all $n+1$ nodes, the interval is partitioned: $a = x_0 < x_1 < cdots < x_n = b$. On each subinterval, a separate polynomial (linear, quadratic, cubic, etc.) is fitted. Piecewise linear interpolation simply connects consecutive nodes with straight lines—trivial to construct but generally non-differentiable at nodes. To enforce smoothness, higher-order piecewise polynomials are needed. A cubic polynomial on $[x_j, x_{j+1}]$ has four free parameters ($a_j x^3 + b_j x^2 + c_j x + d_j$), allowing constraints on both function values and derivatives at the endpoints.

(C) Formal Notations (LaTeX)

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Key formulas

- Cubic polynomial on subinterval: $S_j(x) = a_j x^3 + b_j x^2 + c_j x + d_j$ for $x \in [x_j, x_{j+1}]$
- Alternative form (simplifies calculation): $S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$
- Definitions: S_j is the cubic interpolant on the j-th subinterval; a_j, b_j, c_j, d_j are coefficients determined per subinterval; x_j is the left endpoint of the subinterval.

(D) Critical Connections

- Contrast with global polynomial interpolation: high-order global polynomials can exhibit large oscillations (Runge phenomenon) and non-local sensitivity.
- Hermite polynomial interpolation matches both function values and derivatives at nodes, but requires knowledge of f' at all nodes—piecewise polynomials generalize this idea without demanding derivative data.
- Piecewise linear interpolation is the simplest case but sacrifices differentiability; cubic splines achieve C^2 smoothness without needing f' or f'' data.

(E) Evidence from the transcript

"So they do have their flaws because you cannot just isolate one part of the interval with respect to the other part of the interval."

"why not divide the interval into subintervals? And then you construct an approximating polynomial for each one of the sub-intervals"

"at the nodes, at the interior points, there is no reason why the derivative of this function is equal to the derivative of this function" (piecewise linear case)

Cubic Spline Interpolation

(A) Core Concept & Context

A cubic spline S on $[a, b]$ with nodes $x_0, \ldots, x_n$ is a piecewise cubic function that interpolates f at the nodes and is twice continuously differentiable on $[a, b]$. It resolves the smoothness issue of piecewise linear interpolation by matching not only function values but also first and second derivatives at interior nodes. Humans cannot visually distinguish changes in the third derivative, so cubic splines appear perfectly smooth.

(B) Detailed Explanation

On each subinterval $[x_j, x_{j+1}]$, define $S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$. Impose:

1. **Interpolation**: $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each j.
2. **Continuity of first derivative**: $S_j'(x_{j+1}) = S_{j+1}'(x_{j+1})$ at interior nodes.
3. **Continuity of second derivative**: $S_j''(x_{j+1}) = S_{j+1}''(x_{j+1})$ at interior nodes.

These conditions yield a system of linear equations for the unknown coefficients. However, the system is underdetermined without boundary conditions. Two standard choices:

- **Natural boundary conditions**: $S''(x_0) = S''(x_n) = 0$ (zero curvature at endpoints; corresponds to a free beam in mechanics).
- **Clamped boundary conditions**: $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (derivative specified; corresponds to a cantilevered beam; generalized form does not require f' to be zero).

The resulting linear system is well-conditioned, guaranteeing unique existence of the spline.

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- $S(x) = S_j(x)$ for $x in [x_j, x_{j+1}]$, where $S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$
- Interpolation: $S_j(x_j) = f(x_j)$, $S_j(x_{j+1}) = f(x_{j+1})$
- Smoothness: $S_j'(x_{j+1}) = S_{j+1}'(x_{j+1})$, $S_j''(x_{j+1}) = S_{j+1}''(x_{j+1})$
- Natural BC: $S''(x_0) = 0$, $S''(x_n) = 0$
- Clamped BC: $S'(x_0) = f'(x_0)$, $S'(x_n) = f'(x_n)$
- Oscillation bound: $\int_a^b (S''(x))^2\,dx \le \int_a^b (f''(x))^2\,dx$
- Definitions: S is the cubic spline; S_j is the restriction to $[x_j, x_{j+1}]$; $a_j = f(x_j)$ (from left-hand evaluation); boundary conditions (BC) close the linear system.

(D) Critical Connections

- The progression from piecewise linear (two parameters, two equations per interval) to piecewise quadratic (three parameters, underdetermined without asymmetry) to piecewise cubic (four parameters, four constraints with symmetry) is motivated by the need for smoothness and solvability.
- The formulation $S_j(x) = a_j + b_j(x - x_j) + \cdots$ (instead of $a_j x^3 + cdots$) simplifies calculations because terms vanish when $x = x_j$.
- Mechanical engineering origin: cubic splines model the shape of a thin elastic beam (or ship hull) under minimal bending energy, corresponding to minimizing $int (S'')^2$.

(E) Evidence from the transcript

"I'm going to ask for the derivatives of the interpolants on the left and on the right to be the same"

"you cannot distinguish the third derivative. And this is what the cubic spline does."

"contrary to what happens in regular interpolation...you are bound to bound oscillations of the spline with oscillations of the original functions"

"Everybody should do this once in their life" (constructing a cubic spline by hand)

Numerical Differentiation: Forward Difference and Error Analysis

(A) Core Concept & Context

Numerical differentiation estimates derivatives $f'(x_0)$ from function values when f is available only as a black box (no closed-form expression). The forward difference formula $f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}$ is derived from the first-order Taylor expansion and has $O(h)$ truncation error. However, small h amplifies round-off error due to catastrophic cancellation in the numerator, creating a fundamental tradeoff.

(B) Detailed Explanation

Taylor expansion gives $f(x_0+h) = f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(\xi)$ for some $xi in (x_0, x_0+h)$. Solving for $f'(x_0)$ yields the forward difference formula with truncation error $frac{h}{2} f''(xi) = O(h)$. This is a **first-order, two-point, one-sided** formula.

The **truncation error** decreases linearly with h: $|E_{text{trunc}}| leq frac{1}{2} M h$, where M bounds $|f''|$.

The **round-off error** arises because $f(x_0+h)$ and $f(x_0)$ are represented as $\bar{f}(x_0+h)$ and $\bar{f}(x_0)$ with errors bounded by ϵ_{mach} (machine epsilon). Subtracting nearly equal numbers causes cancellation, and dividing by h magnifies it: $|E_{text{round}}| lesssim frac{2 epsilon_{text{mach}}}{h}$.

Total error: $E_{text{total}} approx frac{1}{2} M h + frac{2 epsilon_{text{mach}}}{h}$. Minimizing with respect to h yields $h_{text{opt}} = 2 sqrt{epsilon_{text{mach}} / M}$.

Practical rule of thumb: for a q-th order method, choose $h gg (epsilon_{text{mach}})^{1/(q+1)}$.

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Forward difference: $f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}$
- Truncation error: $E_{text{trunc}} = frac{h}{2} f''(xi)$, $|E_{\text{trunc}}| \leq \frac{1}{2}Mh$ where $M = \max|f''|$
- Round-off error bound: $|E_{\text{round}}| \leq \frac{2\bar{\epsilon}}{h}$ where $\bar{\epsilon}$ is a function of machine epsilon
- Total error: $|E_{\text{total}}| \leq \frac{1}{2}Mh + \frac{2\bar{\epsilon}}{h}$
- Optimal step size: $h_{\text{opt}} = 2\sqrt{\bar{\epsilon}/M}$
- Definitions: h is step size; $xi in (x_0, x_0+h)$; M bounds the second derivative; $\bar{\epsilon}$ is the round-off unit; q is the order of accuracy (here $q=1$).

(D) Critical Connections

- Taylor series is the natural tool because derivatives are local quantities and Taylor expansions are local approximations with explicit derivative terms.
- Backward difference $f'(x_0) \approx \frac{f(x_0)-f(x_0-h)}{h}$ is analogous but uses the left neighbor; same $O(h)$ accuracy.
- Applications: discretizing differential operators in ODEs and PDEs; estimating derivatives from noisy or tabulated data.
- Polynomial interpolants can also be used to derive finite difference formulas (not covered in this lecture).

(E) Evidence from the transcript

"The obvious thing is we take h small enough and then we calculate this ratio"

"the smaller the h, the better the truncation error is gonna be...But if I have h small, then the two values in the numerator are going to be very similar. And when you subtract them, you're going to have a high probability of cancellation"

"you have a magnification of the cancellation error...because then you are dividing by a very small h"

"There is a theoretical optimal h...which is not always easy to determine"

Comparisons and supporting details

Interpolation Method	Smoothness	Parameters per subinterval	Requires derivative data?	Pros
Piecewise linear	$C^0			
Piecewise quadratic	C^0 or C^1 (if forced)	3	No (but underdetermined without extra condition)	Moderate smoothness
Cubic spline (				
Hermite polynomial	C^1	Varies	Yes (f' at all nodes)	Matches derivatives exactly
Finite Difference Formula	**Type**	**Accuracy**	**Truncation Error**	**Typical Use**
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Forward difference	One-sided, two-point	$O(h)$	$frac{h}{2} f	
Backward difference	One-sided, two-point	$O(h)$	$\frac{h}{2} f''(\xi)$	Left boundary conditions

Action Items

- [ ] Complete the cubic spline construction example with three points $(1, 2), (2, 3), (3, 5)$ using natural boundary conditions (homework assignment)
- [ ] Work through the numerical differentiation example: estimate when to stop decreasing h when calculating $f'(1/2)$ for $f(x) = \sin(x)$ with tolerance $\frac{1}{2} \times 10^{-6}$ and $h = 4^{-n}$ (to be continued next lecture)

Notes

Transcript

So they do have their flaws because you cannot just isolate one part of the interval with respect to the other part of the interval.

So that might be considered a flaw, let's say. So people start thinking about why not, instead of considering just one interval, So one polynomial for the whole, for the full interval where you are interpolating, why not divide the interval into subintervals? And then you construct an approximating polynomial for each one of the sub-intervals that you are considering. Okay, so this is called piecewise polynomial interpolation.

So this is one answer to the many flaws that you can have. when you're just doing polynomial.

And the basic approaches are, well, piecewise linear interpolation.

So you join the points. by straight lines. So these points that you have here are the nodes of your interpulent. And instead of having one high order polynomial going through all these points, what you have are first order polynomials going in between. between each one of the points. This is very, very easy to construct.

The disadvantage is that you have no, in general, you have no differentiability at the nodes, right? Because if you are, so the picture, I think it's much easier than my words. So this is the interval of interest, right? A, B here. And then you have, so this is in general your X naught. And your B is your X n. And then you would have here x1, x2, dot, dot, dot, dot, dot, all the way to xn minus 1. So these are the nodes where you have the values of the functions that you are trying to interpret, right?

So in general, if you're doing an interpolation, you would have something like this. But now if you're doing piecewise linear interpolation, this means that you would have here And here you would have the interpulent would be just a line because it's piecewise linear. So you just connect the dots with a line. Like this. And the problem or a disadvantage or something that you need to be aware of is that at the nodes, at the interior points, there is no reason why the derivative of this function is equal to the derivative of this function.

So you might have non-differentiability at the nodes. And this is what I'm stating here. in the second statement.

Of course, you can go ahead and just for each one of the subintervals, you can instead of choosing just an interpolation, you can come up with a Hermite polynomial where you not only ask for the value of the function to be in the interior points to be equal to the value of the interpulent, but also the derivatives, right? This is the difference between polynomial interpolation and Hermite polynomial interpolation.

But then you have the extra cost of having to come up with f prime at these endpoints of the subintervals. So this is one disadvantage, this is another disadvantage. People start thinking, "Okay, so maybe I can do better." So the story is I have AB. I'm going to divide AB into subintervals, and then I'm going to approximate-In each one of the sub-intervals, I'm going to approximate my function. I'm going to interpolate my function.

So why not? So my problem is maybe linear is not good enough. Why won't I go and just maybe ask for a quadratic approximation?

So I go and I say, let me see. Before here. Just come up. Paige. So what I'm saying is why not Why not? Why not? AX squared. plus BX plus C. in each sub-interval. So what I'm saying is, again, I have my A, I have my B. I have here x1, x2, blah, blah, blah, blah, blah. X and minus 1. Minister Asking. for just the line ax plus b, I'm going to ask for a parabola. Bye. And then if you ask for a parabola, Do you understand what I'm saying?

So here, I was just asking for a line. And here I can, I'm asking, can I do better? Can I just ask for a problem? And then if I ask for a parabola, I have three parameters that I need to solve for. These parameters. Okay. And if I am at an interior node, let's say here, Let's say I'm considering this sub-interval. At this point, I'm going to ask my ax1 squared plus bx1 plus c to be equal to the value of the function at x1.

So this is the interpolation. Right on the left hand side Note in the sub-interval, so this would be here. So this star corresponds to this point. And then if I ask for interpolation at v, Right hand endpoint. So to start, I'm going to have ax2 squared plus bx2 squared plus c equals to the value of the function at 2. Right? So this is the interpolating. idea.

Remember, you're interpolating polynomial, which is this guy, in this interval, because it's interpolating polynomial, by definition, it has to be, at those nodes, it has to be equal to the value of the function that you're trying to interpolate. But you see here, you have two equations. for three parameters.

A, B, and C. So my parameters are A, B, and C. Right? And here I have just two equations. So maybe, so this is the question, maybe I can ask for one more question. 14, right? So I would have Three equations. that would determine the three parameters that I'm looking for. So maybe I can ask.

Four. Ah, the derivative? At this point, it would be the same. or the derivative at this point to be the same, but then if you just, because you just need one more equation. But if you ask For the derivative on the left or on the right, you would

have some sort of a symmetry between what you're asking from the left-hand point from what you're asking to the right hand point. So what makes sense is actually to ask for the value of the function and the derivative here and here to be the same.

So you would have four equations. Right? So two equations for f, two equations for f prime, And then you would have, Four equations. Right, two plus two to four. And this means that we would be able to determine for... parameters So you can actually go cubic.

Do you understand what I'm saying? So I'm just thinking about... The story is... I can go and do me here.

Linear. But clean air may not be good enough. I can maybe ask a little bit for more. Why is it not good enough? Because you might be too discontinuous at the notes.

because you would have a smoothness problem at those particular nodes.

So people start thinking, oh, maybe I can do better if instead of asking for a linear approximation, I'm going to ask for a quadratic approximation. But if I ask for a quadratic approximation, then I just have two equations and I have one degree of freedom because I have three parameters, A, B, and C. So I'm going to ask for more. And then I say, OK, but if I ask for the derivative to be the same at x1-I cannot ask for the derivative to be the same as x2 because then I would have-for equations for three parameters, right? So this is too much. So I'm going to ask for a cubic interpolation.

Because if I have a cubic interpolation, I'm going to have four parameters before you wait. And this is how splines came along. So this is cubic spline pervulation. You have cubic polynomials in each one of the sub-intervals. So you have four constants to be determined. the A, the B, the C, and the D, right? And then I'm going to use this constant, or I'm going to find out this constant, A, B, and C, and D, to actually impose some sort of smoothness in the interval.

So I have one interval, and I'm going to use what's happening at the end points of each one of those subintervals Ask for the smoothness.

of the final interpulent spline, yes.

So if you look back to the No, no, no, no, no, no. The derivative has to be the same as the derivative of x1.

The derivative of the intercalant and the function have to be the same. for example. But this is not what I'm asking for a cubic spline. Because, thank you

for asking, I'm not going to require to know more about the function. Because if I require f prime to be known, then I'm back to-Least flaw. where I need to ask what is the derivative of f prime. What I'm gonna do is I'm gonna ask for the derivatives of the interpulance on the left and on the right of the node to be the same.

So until now... When I say I'm going to ask for the derivative,Dean Turculan. to be equal to the derivative of the function that I'm-trying to interpolate, this would be Hermite Polynom. Right? Because The derivative of the turbulent is equal to the derivativeHave the function. But this is not what the cubic spline does. A cubic's flying asFour. So you let me like give you the picture.

So I'm here. Let's say this is a node Xi. like an interior node, right? Here. So, So this would be here. F of X, I.

And then this node is a node that corresponds to this interpolant and this interpolant.

Right? Because I have here xi minus 1, and I have here xi plus 1.

And in each one of these intervals, I'm going to interpolate. Right? So whatever happens, the cubic function that I have on the left is going to be different from the cubic function that I have on the right. Because remember, the idea of piecewise interpolation is to, instead of interpolate, instead of using the same polynomial in one, I'm going to use one interculent in each one of the subintervals.

But remember, what I want, I want smoothness here. I mean, in every interior note. So if I want smoothness, I'm going to ask for this interpolant here. to be interpulent at this point, which means that This function, this intercalant function at this node is going to have the same value than being turbulent on the right hand side. which is equal to f of xi because they are both intercalates. But then after that, I'm going to ask for the derivative of this guy to be equal to the derivative of this guy at this point.

And more, I'm going to ask for the second derivative to be the same. So visually, When you, when you, when you like draw all the turbulence The only difference at The note is going to be in the third derivative. But we as humans, we cannot distinguish This change in the third derivative of a function.

You can distinguish the first derivative where you can see that the function has different slopes.

You can distinguish concavity, which corresponds to the second derivative. but you cannot distinguish the third derivative. And this is what the cubic spline

does. So the difference is I'm not asking for the derivative of the spline to be equal to the derivative of the function. I am asking for the derivatives of the intercalants on the left and on the right to be the same. which is different. So this is not hermite polynomial interpolation.

Okay, so, woo!

So this is the definition of the spine. The cubic spline is very, very famous. It started, you will see, with mechanical engineering. So all of this because mechanical engineering people were trying to...

I think it has to do with the ship, I think, if I'm not mistaken, if I recollect well. I think they were trying to smooth out the hull. Hull? Yeah, like the shape, like the thing where, It's a whole other shape. Bye.

So they were trying to make it smooth. So you have a function f defined on ab. And you have nodes, x0 all the way to xn, so this is the same notation that we have been using so far. And a cubic spline in turbulent S For F, so S is the intercalant and it's defined piecewise. So in each one of the subintervals, the A, the B, the C, and the C are going to be different. And how do I define, how do I determine those?

D. Let me just point out, the function itself is s. So this is my function. And the value of the function is S of G, where S of G is the cubic polynomial on an interval that is defined by XG plus 1.

So you would have your spline here. Some more here. And then... This would be the S. This would be your x. And this is xj-j or j?-xj plus 1. So... Here, the function is sj. Is this clear? So the notation can be a little bit... It's harder to follow. And then because it's interpulent, the value of the function at x_j has to be equal to the value of the function at x_j and on the right as well. So here and here The function sj at the nodes, at each one of the nodes of the interval, in this case I only have two, then what happens is this value is the same as the function and this value is the same as the function.

So we interpolate the points. Okay, but you can see that You might think-so this is important to count the number of equations that I have. that I would have for each one of the intervals. I would have two equations. So this first equation for interpolation and this second equation for interpolation. But I do not have...

because If you think about the next interval, so this would be S of J plus 1. And here I would have S, the X of J plus 2.

But when I ask, For interpolation at this one point, x of j plus 1, I only have one equation because the one coming from sj is the same as coming from sj plus 1.

So...

When you state these first two equations for the interval corresponding to sj, this implies something about what's going to happen on the left side of the next interval.

So this implies what you have here. So you don't have as many equations as you expect.

And then the same thing can be said for the first derivative and the second derivative.

Okay. Did you understand what I just did? Did you understand? So let me just refresh. This is just the definition have the function spline.

It's defined differently for each one of the intervals and this Sj is a cubic polynomial. So you would have...

that xj of x is going to be a, I think I did this, a j x cubed plus, bj x squared plus cj x.

plus DJ So each one of the parameters depends on the interval that you are considering. Okay. Is this okay? Now, because The xj's are uterculents. They have to be... Yet the nodes, they have to be equal to the value of the function at the points.

This is by definition of interpolate.

And they have to be an interval on the left.

Hand point, XJ, and on the right, XJ.

But because What's happening on the right of SJ is what's happening on the left of sj plus 1, because they glue together. This is already included in the previous equation, so you cannot double count the equations. And then you do the same thing for the first derivative for smoothness and for the second derivative also for smoothness. But if so happens, then you still don't have-with all of this, so from 2 to 3, that's still not enough equations to come up with the number of parameters that you have in the cubic spline.

So we usually choose One of two sets of boundary conditions. The first one we call natural free boundary. So if you are a mechanical engineer, you would recognize this. That has to do with the second derivative being equal to zero at the end point. So this is the natural cubic spline by definition. Or, so you either choose this one or you choose this one. You have a clamp boundary. where the first derivative is equal to the value of the function.

is equal to the derivative of the function at those points. There is another definition of a clandestine boundary. Where you require this to be equal to zero. Thank you. For us, it's going to be. Yes.

This one?

So this is where you So this would be where the moment would be equal to zero. So the curvature of the-I don't know, b or whatever is equal to zero.

This is easy, right?

This would correspond to what we call a clamped, clamped?

Cantilevered. Cantilevered kind of beam, sorry. I am a mechanical engineer, but I forget the terms.

But in mechanical engineering you would require if it's cantilevered on both sides this would be zero.

So there's no curvature, right? So this is what happens.

But for us, I mean, for math people, there's a more generalized form where you just request the first derivative to be equal to the first derivative of the function. And those, this, this two types of boundary conditions are very well known.

And there's a lot of theorems about Uniqueness and existence. of So of the cubic spline, for these longer conditions. It has to do with how you write the system of equations.

So this is the definition. So if you want to come up with a cubic spline, You have to write down all the equations. Those are linear equations. Linear, you determine the parameters, a_j, b_j, s_j, and d_j.

And you just solve those equations for-for the cubic spline that you are looking for. It's um... So what I did here, I'm going to leave this as homework.

You have to do it once, and I will tell you what the trick is. And once you have the trick, the calculations are not as bad as they seem, but you have to know the trick. Okay, so I think those are the points that we had before, 1, 2, 2, 3, and 3, 5. And I want to calculate the natural. So the natural cubic spline is the one with... 5.1 boundary conditions. So then the cubic spline is defined. And you can see that you have, so you have two subintervals from one to two, and then from two to three. So you would have an S1.

This is this guy here.

And then you would have an S2 corresponding to this guy here. So it's a piecewise defined function.

And so How do you get this? How do you do this? So you go back to the equations and you just write them for each one of the subintervals. And the trick that you should, instead of writing C, I wrote them like this, AJX cubed plus BJX squared plus SJX plus DJ. This is the way you think about the cubic interpolation in each one of the subintervals.

But the way to actually calculate the SJ, S1 and S2, in the case of the example, is to write the SJ in terms, of the left hand and point. So you write aj plus bj, x minus xj.

plus cjn. If you do this, the calculations are much easier because a lot of the equations get to be like these last three terms are going to be equal to zero when the x is equal to xj and that simplifies the calculations. So, but everybody should do this once in their life. And you can see, I was telling you this is, I mean, it's a very smooth curve. So here at the-The only interior point that I have, because 1 and 3 are the a and the b.

So at this point, at x1, let's say, you see that-I mean, the curve, the graph of that function looks very smooth. because the first and the second derivatives on the left and on the right are equal. So this is the idea of this poem. of a cubic spline.

Now, I kind of wrote here what you should do. I think you have to do it once in your life.

I mean, if you do it just with three points, it's not a big deal.

but you still have to choose four plus four. So you still have four equations because you have one equation for each one of the parameters. So one, two, three. Three four 5 and then you would have 6 that is going to be equal to 0 corresponding to x minus 1 squared.

And then, um, Sprite 670.

So you have-Eight equations, linear equations that you can easily solve.

OK, and the unknowns in the linear equations are just the parameters for--Where is it?

R to A, 1, B, 1, C, 1, B, 1, and then A, 2, B, 2, C, 2, and A. Okay, so once you get those, you would have...

Like in a compact form. You have These equations The AJs are known. Why are they known?

Because By definition, On the left hand side, all the terms are going to be equal to zero. Zero. So the aj's are the values of the function at xj.

You have to calculate that. This is not actually a very smart way of doing it.

So the unknowns are actually just The, um, the CJs.

because I solve for the beast.

So if you put this equation, this is just one of them, one of the A, if you put them into a AX equals B kind of format for a linear system of equations, this is how you discern it. the solutions and then you look at the system, I mean you look at the A, you look at the rank and what not, And there are two theorems that show that the natural cubic spline exists.

So If you have the interval-Then F has a unique natural cubic spline. On the notes, so this is the theory, but this is all about linear algebra, it's not very hard to show, and you have a similar theorem for the clang cubic spline.

So if you take, there are other cubic splines that are not natural or clanged. But these are the most common, and because they are the most common, they are the ones that we actually know a lot about.

And if you put it together with just what we know about solving linear systems, it's very easy to show that the solution for the system exists and is a unique solution. I'm not going to go into my community at all.

But this is only to show that the spline exists and is unique. Okay, so one big problem that people were trying to solve is, if you remember what I said in the beginning was one big flaw or one flaw of regular polynomial interpolation is that if you have an oscillation in one part or a fluctuation in one part of the interval a, b, Then you're going to have high oscillation because you increase the order of the polynomial.

You're going to have higher oscillations everywhere in the interval because it's a high order polynomial.

This does not happen. with cubic splines. In fact, you can show, and I've written it down here in terms of the second derivative, and here the second derivative is a measure of the oscillations of, well, second derivative squared is a measure of, the oscillations, local oscillations, so when you integrate it out, you would have The integral of AB from A to B of the second derivative of the cubic spline squared So this is a total measure of the oscillation of how many oscillations and the degree of the oscillations for the cubic spline.

And it can be shown-that this is less is bounded by the oscillations in the function itself. So you contrary to what happens in regular interpolation, so polynomial interpolation, if you're making a cubic spline, you are bound to bound

oscillations of the spline with oscillations of the original functions or of the original data points that you have. So yeah, so people love cubic splines.

So this is, where I leave interpolation, and this is how far we're gonna go into interpolation.

Our exam.

So next topic.

Next stop, yes, questions? So why the smiles? No smiles.

No, we are sad. We are sad now.

It's actually my favorite topic. I don't know why, but I always liked numerical differentiation much more than numerical integration.

I don't know why. I think I like the calculations, like the stupid calculations that I can do with numerical differentiation. And, which are more or less the same, the ones that I can do with numerical integration, but because I usually learn differentiation, I think this is where I got my ideas of BALM stuff and whatnot. So this is why I like it.

So the goal in numerical differentiation is The differentiation is basically we want formulas to calculate derivatives.

And we want to have an idea of what the error For those formulas,Is or are?

I'm going to use Taylor polynomial, and I'm going to use polynomial interpolants as well. But for now, Taylor polynomials is the obvious thing to do because if you're calculating a derivative, Then this is a local entity and when you're doing Taylor approximations or Taylor polynomials, they are also local and they have a whole bunch of derivatives in them.

So it's the natural tool to come up with.

And then there's a great thing which is called, which is turning into extrapolation where you, it's an art, it's an art, where you compute high order derivatives by-so high order formulas for derivatives from formulas of lower order derivatives.

It's like Tetris, you know the game? but for numerical differentiation.

It's one of those things that you think, they go like, "Ah, who would come up with this?" But Richard Toussaint, he did. So, but first today, my goal is for you guys to appreciate how Taylor changed the world. And so the idea is that we have a black box. So this is my function f. We can Evaluate the F so we know the value of the function endpoints, but we don't have a closed formula or a closed form

expression, algebraic expression, that for F, that you can use to differentiate, right?

Because if you had, then you would just...

use what we know about derivatives and you would just differentiate the expression for the function. So for example, if I know that f is equal to, I don't know, sine, right, then I don't need numerical differentiation because I have the formula for f.

So if I want to differentiate the f, what I'm going to do is I'm just going to differentiate the sine. I'm going to get the sine. But this is not the problem that I'm trying to solve here. The problem is I can't calculate f. but I don't have the expression for f. So I can calculate the value of the function at each one of the points. But my goal is to use the fact that I know the value of the function at the points to extrapolate for f prime, for the derivative.

Okay, so this is what we have and our goal is to compute approximations of f prime, f prime, all the way to how many derivatives you want at the nermitary point in something that I hope to be A point x naught, yes, I'm calling it x naught, happens to be in the domain of f of f prime and whatnot. And so this is the easy part. Then I want the hard part as well, which is to estimate the error that I'm making.

And you will see that I'm very interested numerically stable AAM methods.

So you will see that a big problem when you are differentiating.

I've told you about this when we were discussing at the beginning of the class. Of course, we were discussing how round-off error keeps creeping in, in particular cancellation, where you have to subtract two values that are very similar. This is what happens quite a lot. in numerical differentiation. So this is something that we We want to come up with a balance on the errors, not only on the truncation error that comes from Taylor, but also to make sure that we have a minimum value.

the possible minimum value for the round of error. when we're making those subtractions.

So let me be a little bit more specific. Oh, first of all, why are people very interested in numerical differentiation? Well, because they want to solve ODE's and PDE's. So we want to approximate the derivatives that appear in these ODEs and PDEs, also integral equations.

But I mean, what we usually are looking at is a differential equation.

So you just need to come up with a discretized version of the differential operator independently of the order of the operator.

And then of course have some sort of bound on the error that we are making in that discretization.

We also need to come up with, this is, Is...

Maybe a bit of a hot topic right now.

We want to come up with an estimation for the derivatives based on data points.

So again, based on the F, not the F prime that we usually do not have.

There are other strategies, I'm not going to talk about them, to actually come up with numerical differentiation.

This was a hot topic when I was, I think, doing my masters. I don't remember, I think my masters, yes. which is automatic differentiation, we will not cover this here. Okay, so we will not cover this in the course.

Okay, so Easiest thing ever.

Let's say I want to come up with a little formula to calculate the first derivative of a function at the point x naught. So everybody remembers the definition of the derivative.

So it's a limit when h approaches the zero at x naught plus h minus the value of the function at that point. all divided by h. So what we usually do is, the obvious thing is we take h small enough and then we calculate this ratio, right?

So you take the subtraction and you divide by h. and you will see that the truncation error you will see that I'm gonna use Taylor series.

The truncation error to calculate the derivative decreases Wow.

faster or not, it depends on the order of the method.

As my A shrinks, But this is true as long as F is smooth enough. I've written down F is smooth enough and then I define F is sufficiently smooth if the high order derivatives exist and are bounded because this is what we will need to calculate errors. But the problem there, with expressions like this one.

So this is just to calculate the first order derivative, right? There's a problem.

I mean, there's two problems. One is obvious. You have the So let me go back.

The smaller the age, the better the location error is gonna be. because you remember that the error in Taylor, I will show you that, is proportional to a power of the age.

Um... So on the one side, I want h small. But if I have h small, then the two values in the numerator are going to be very similar.

And when you subtract them, you're going to have a high probability of cancellation.

So your round of error is going to increase, but it gets even worse because then you are dividing by a very small error.

So you have a magnification of the cancellation error that I was mentioning before.

So you have two conflicting points here. The first one-so this is-like a common factor to any numerical differentiation scheme. On one side, because you're going to be using Taylor series, you're going to be your age small enough. On the other side, you don't want it to be too small because if it's too small, You're going to have cancellation and you're going to have magnification of the cancellation so your round of error is going to be too big so you have to kind of Have some sort of idea of how to balance These two conflicting roles. So this is.

Maybe the thing that people do most in numerical differentiation. So I will-I'm gonna use stellar approximation and you will see how natural this comes.

So this is the first rule that everybody learns. It's called, well, either the forward difference rule or the backward difference rule.

It depends on how you think about it, but you can start from the beginning. So I'm going to consider the first order Taylor approximation about x naught.

So this means that I have the Taylor polynomial up to here.

I'm going to have here, this is oops. This is my Taylor polynomial up to first order, right? Because I have here the first derivative of the function. I am taking x naught as thePoint. central, let's say, to the neighborhood that I'm trying to make the Taylor expansion about.

And then you have here the error corresponding to the truncation.

And you guys remember that my epsilon, my psi, sorry, my psi is a function of the x. and the side stays in between x and x-node. This is assuming-So x naught and x. X naught and X. This is assuming that my age is larger than zero. change larger than zero. If you assume, so basically you're here, this is your x-naught, and this is your x, and this guy here is h larger than 0. Okay, so you are coming up with a value of the derivative of the function at this point using two points.

This point, x-naught and x. So you rewriteA full Taylor, polynomial of the first order. in terms not of x and x not, but in terms of x not and h, by substituting

your x by x not plus h. All right, so what I hear was, I said this guy is x naught plus h. And then this is X naught plus H. The X naught cancels with the X naught. And then you just have the H. And then this part corresponds to something that is big O of H squared because this is here H squared.

So this formula and this formula are the same. Okay, one is written in terms of x and the other is written in terms of the age, but it's the same formula.

And then what I do, remember, what I want to know is to calculate, or I want the formula to calculate f prime.

So I'm going to solve this equation for f prime.

And then when I do this, I have to divide by h. So my big O of h squared becomes, when I divide by h, becomes big O of h only. So you have here your first Um. Your first way to come up with the derivative of the function at the point And we say this is a first order term, a first order scheme or rule. Because You have O of H1. So if you had O of H-Q, I think I'm using Q. Let me see. Sometimes I use R. Okay.

Let's see if you have O of HQ. then this would be a Q-th. So this is called the accuracy of the method. And you can see that the truncation error, because it's O of H, it decreases linearly with H.

So this is a Forward difference rule, if you choose to do the same thing but with your x on the left of x naught, then your h would be negative and we have a... a rule that you call the backward difference rule. So with the forward difference rule, graphically what we're doing is we're calculating the slope. to the right and then if you do a backward difference you're calculating the slope to the left Okay, so these two rules, Because they require two points.

We have the x and the x plus h.

They are said to be two point formulas. So I want to calculate the derivative at one point. I am using the value of the function at two points, at x plus h and at x naught, and because of this, I am... using two points and because of this it's a two point formula. They are one sided. Because if I want to know the value of the derivative x0 with the forward difference, I'm going to look And what's happening on the right-hand side of x naught, because I'm using x naught plus h.

So it's a one-sided kind of rule.

It uses the value of the function at the point and the value of the function at another point on the right. The backward difference uses the x naught, the value

of the function at x naught, and also the value of the function at x naught minus r, plus h, but the h is negative.

So again, in one slide. And you have a truncation error FOH, and this is why it's called a first order. or scheme.

Okay, so Let's say I want to come. I want to come up with a numerical approximation to f prime. At one point two. and my function is actually the sine of x.

Of course, this is a stupid question, right? You don't actually have the value of the function. We don't have the algebraic formula for the function.

But I know that because I know how to differentiate, that what I'm trying to do is I'm trying to approximate F prime, which is the cosine. at 1.2 and this is the discoverer. And if you apply, in this case I applied the forward difference rule, What I did here to come up with this nice little table was I wrote different values of H. So from point one all the way to 10 to the negative seven, Using that forward differential And I calculated the absolute error that I, because I can't, why can't I calculate the absolute error?

Because I have the value that I'm approximating, which is just the cosine of 1.2.

So this is here. So I have The difference, so f prime at x naught, so this is the forward. Difference rule, which is the rule that I used, is going to be equal to, I'm just copying this, so the value of the function at x0 plus h, So x0 plus h.

In this case, my x0 is going to be equal to 1.2 because this is where I want to know the derivative.

So this guy is my x0. minus the value of the function at x zero, this I can calculate, divided by my h.

And then for different values of h, I've calculated the error.

What is the error? Well, if this is the numerical differentiation of the function at x naught, the absolute error is gonna be an absolute value of the numerical differentiation scheme minus the value of the function at the point. And of course I know that this is my cosine of 1.2. Maybe I can do it for you. Minus cosine, well... My x-naught is 1.2 cosine of 1.2. So this is the absolute error that I'm calculating here.

Is this clear what I'm doing? And you can see That, well, it's here. That the error is doing exactly what I predicted to do is every time I get, well, One obvious thing is that the error becomes smaller and smaller as my age becomes smaller and smaller.

This is the obvious thing.

But what I want you guys to see is that it behaves linearly.

Approximately linear. So I have For age equals to 0.1, I have There you stand to the negative.

Choo.

And then 0.01 approximates with 0.10 to the negative 3 and so on.

Okay, so this is because the method is O of H.

So the error is going to be linear.

Okay, but then... And the surprises appear. So let me go back.

to what I was just saying. When I write...

When I write this here, This is just a truncation error. Let me just write it down. It has nothing to do with the round-off.

So now no round off. Included.

And why is this relevant? Because this is just a farPart of the story, Okay, I'm gonna happily going the error comes smaller and smaller. It's linear in the age.

I'm gonna go and And I continue to decrease the value of H.

And then at the certain point, you see that for h equals to 10 to the negative a, It goes to 10 to the negative 10. which is more or less what we were kind of doing before. And then it starts to increase. So you see, I would expect the, if you're doing just location error, you would see the same trend as we had before, but not, this is not true. And this is the round of error kicking in. So This is why I say here that these results especially when you see an increase of the error with the decrease in the value of h, has to do with the rounding error coming in.

Okay, so... In a first analysis, we'd say that truncation error decreases with decreasing age, but the round of error increases with decreasing age. And you can actually do the calculations. This is just a graph of what's happening. So what do I have here?

I have in the dashed red line-So this is log log. Okay, so here you would have on the horizontal axis, you have the H. So this is just... The blue grass is what I had before. In like in table one. Because you would have here on the vertical axis, you would have the absolute error.

And then.

The blue line is I start to decrease The value of H.

Going this way, right? So I start at one.

I start to decrease, and you can see that the blue line is linear. And then at a certain point, there is a certain value where it starts to increase. So the first part, so this means that the first behavior or the first regimen corresponds to the truncation error. Being The error that you're looking And then when you start to see the increasing in the second part, you will have your rounding of error being larger than the value of the jolgation error.

So this is about the blue light. The blue light. The blue light. About the red dashed line, what you have here is-The truncation error without looking at just the, without seeing the round of error. And then the question is, why is this linear?

because So this is the dashed line. The slope is, oh, it's more than linear. The slope is one. Because if you do, so if I want to answer this, Maybe I can answer here. Here, so this was the question. Why is the slope One, equal to one. So the error was, let me, The location error, I think I usually. So this is location error. So it was in the case that we were studying, it was one half. of f double prime psi, right?

multiplied by h squared. This is the error corresponding to the method forward difference. becauseUh. You can see here, multiplied by h, so just by h, just by h. So O of h, because then I would have 2 log of h. Yes, so it has to be just... So then... If you ask why is this slow? One, so if you apply the law. If you apply.

So if you apply the log of the absolute error that you have, this is going to be equal to the log of 1 half of f double prime psi multiplied by h, and this is just log of, well, 1/2 f double prime x i plus log multiply. by H. Well, h here is positive, but I'm just keeping the absolute values.

So this is what we have on the x-axis, right?

So you see that it's multiplied by 1. And this is why the slope is equal to 1. If I had If you had it method of another order, Right, so you would have-something, something, you will have something here, multiplying h to the power q. When you did the same thing that I did here, you would have Here, h to the power q, whatever you have here, it would be something different. It would have to be another derivative of the function.

right, then you would have log of h to the power q that you then can put it down, right, outside the log, and you would have the slope corresponding to the q. You see how, how, how. So this is why you have the slope equals 1. Need a log log.

Okay. So, um... Okay, so this is the story. I'm gonna do the calculations for you because we have to see it once.

So round of error. For a very small age, And sufficiently smooth air.

So this means that I can bound the derivatives that I need to bound, the derivatives of f that I need to bound. I will show you how to do this. We may encounter severe cancellation amplified by the h minus 1 that we are multiplying. Did the death cancellation happen?

As I mentioned before, this is a common problem not just for this particular scheme, And on the one hand, you want to use a sufficiently larger age because I don't want my results to be affected by the round of error. But on the other hand, you want the smaller age so that the truncation error gets smaller, okay? So a way to do this-so did you understand this is like the main issue of this class? But then you can do the calculation.

So I'm going to calculateUm.

First I'm going to calculate the the bound, I'm not gonna calculate the error, I'm gonna calculate the bound on the truncation mirror.

Sorry, this we already have. I'm going to calculate a bound on the round of error.

So, bounce.

Found on truncation area.

Why am I always doing, okay, so I can do okay. Life wants me to start with location error. For this case, it's very easy. So this is easy. Because the truncation error was for this particular scheme. So I'm doing this for Forward. Differences.

So the error is what you left outF. the Taylor expansion, right?

So it's going to be one half in this particular case, so this is an H. One half of f double prime of xi multiplied by h. Okay, so this is easy if you want to bound it. then this is going to be I'm going to assume that I have a bound on the side. It's an amp. And in this case, for this particular function, because my f was a sine, I know that the bound on the second derivative is going to be a 1. because either the sine or the cosine, and this is true for any derivative of a trigonometric function.

This is going to be one. But in general, I don't have to. So I'm going to keep the myM. And then I multiply by H. So this is the easy part. The second part, which is a bound, On the round off. So my roundOkay, so let me do notation first. So I'm going to think of D, F.

and df bar, maybe bar first, df bar, and df. So this is the value of the derivative With F. So this is just the formula. So this means that my F, my F bar of whatever is a The floating point representation of FIt's whatever. So... Well, I should actually put here also a floating point, but this is beside the point.

So this first expression that I have here is me using the The forward differences I'm calculating the derivative of the function f X naught including So this is just including the round of error. This here is the exact Expression for duration.

So... This is what we actually wanted to calculate. So here This one includes So this is Formula? for f prime with Round off.

This is the exact formula for f prime.

Do you understand the difference?

So my F prime Sorry, my F. bar is going to be equal to the real value of f Close the error. EFR, so this is the one we have. And the round of air. If you guys remember, he's bounded by-a function that I'm calling an epsilon of the rounding unit.

So this is here, the machine epsilon.

So I have a floating point representation of the value of the function. And the difference between both of them is what I hold around me out there. And then this bounding of error is bounded by an epsilon of, what is this called, mu, which is a machine epsilon. that you have seen before. And now what I'm gonna do is, I'm gonna countPopulate. Ah...

The Amber The round of error,Committing when ICalculate.

The round-off version instead of the exact version. This is okay? Do you understand what did I wrote everything that I wanted? So do you understand the difference? So first part is first. Do you understand this bound on the trocation error?

So it's going to be, for this particular case, it's going to be, because I'm looking at...

Forget that I know what my f is. If I know what my f is, I can calculate the m, which in my case, for this particular case, when my f was equal to sine, was just 1. Right, so verify. If I want to take things a little bit more abstract, and if I'm looking at forward differences, I will always have an arrow like this.

So the bounds, or the absolute value of my truncation area is just one half of m. Multiply by age, I'm assuming that my age is positive.

So this part is okay? So this part here has only to do with Taylor, Taylor expansion. This other part is not so easy, but it's me calculating what is the error that I'm making when I'm rounding off. So this is what I wanted to calculate.

You didn't like... In... in...

If I have no round-off error, I would just calculate the value of the func-sorry. the value of the function at x plus h subtract the value of the function at x naught divided by h and I would have the exact formulaFour F-bomb. But what I'm saying is, because we have round of error, The exact formula is going to be polluted by a round of error.

And I want to have an idea of how much that is. So this is the calculations that I have.

And to do that, I'm saying that-The difference between the F, the exact F, which is the guy without the bar, And the round of error version. of f is just this E of r that is bounded by my subfunction of the machine epsilon, the round-off unit that I'm working with.

So what I'm going to do is I'm going to calculate the F bar. X naught minus So this is going to be-I'm just calculating. Here, I'm just plugging in the values. So I need nothing here.

Right? I did not make it. But this is going to be equal to...

So now what I do is I'm going to compare this guy which is the version, round-up version of this one. And F bar minus F divided by H is bounded by this error. Two, three, four.

The less than or equal to The roundup error at x-minus h plus h over h minus-The round of error of f at x_0 divided by h. So this is less than or equal to... Well, of course I have an H. And I also have Oh, let me, maybe I can do this shorter. So this is triangular inequality. So this is going to be equal to the sum of the absolute value of both. Try triangle. And this is less than or equal to. Two times over h. multiplied by the x1.

a function of theSo this is about...

on the round of error for my little formula to calculate the four differences. Okay, so So this is the hardest step. The last step, which is the most interesting step is, Remember what my problem is. I have two sources of error. So if I wanted to calculate, so this is part three. If I want to calculate the total error, I'm gonna have I'm going to write it in English.

The truncation error. Plus... The Rob of Terror.

You can see already that the truncation error goes with H.

We've calculated before. It goes with age. Where is it? Here, it goes with H.

So the smaller the age, the smaller the truncation error. But the round of air goes with 1 over h. So the smaller the age, the larger the rondo hair. So this is-this are the kind calculations for the things that I have been telling you.

that the location error goes with age or with the power of age. It depends on the order of the method that I'm considering.

Whereas this guy usually knows we've run over h. So, you have these two conflicting? Age values that you want to sort out. Okay? So what do I do? What is the sweet age that I need? Right? On the one hand, I want a small guy. On the other hand, not small enough. So what we usually do, and I'm doing this for four differences, but you can actually do it for anything.

You come up with a formula For your error, That depends on...

That depends on age. So the one is one half of mH, and the other one was... 2 over h.

x1 multiplied by Right.

And now, What do I want? My goal in life is to minimize the error.

Minimize the error. So if I want to minimize the error, what I'm going to do is I'm going to differentiate my error with respect to h. And I'm going to set it to zero. So I have a formula for the error. Here's my formula. I know how to differentiate. I know how to make a derivative being equal to 0. Okay, so I'm gonna find This is what I'm going to do. So I'm differentiating this. So I differentiate. I'm going to have 1/2 of m.

minus 2 over h minimizer. Excellent. Yes. equal to zero. And then what I do is I have a gene.

is equal to four Epsilon, bar epsilon, divided by m. Did I do it correctly? Square, I multiply, did I do it?

So I multiply, I'm trying to, so I just multiply this by h. It's easier just to do it other...

But I'm... I... I... Okay. So.

1/2 of m, a, means minus 2. This guy is equal to 0. And then-oh, no, I had it right.

So this guy goes to the other side. He gets to be 4. I multiply 2 by 2. Awesome. Divide by.

Okay, so. The point is This is my point. The point is, You have A value Four. The age that minimizes the error. And this error has two parts: the truncation error and the round-off error.

Okay, so this is theoretically, of course.

And you have here The picture. for what I was saying. So there is a theoretical optimal age, the one that I just calculated. Which is not always easy to determine. So in this case, it was very easy to determine. So in the graph that you see, You have in the green, the ideal truncation error. And in the pink, you have the ideal roundup error. And both of them do what they're supposed to be doing. So the ideal truncation error decreases with the value of h. This is the green thing.

Where is the pink? A dashed line corresponding to the ideal round of error increases with age. So what you want is, because on the left-hand side, on the vertical axis, you have the absolute error. What you want is The value of H that minimizes the error, which corresponds to the point or to the age where both Errors intersect where both lines intersect.

So corresponds to this case and to the negative one.

Okay, soReady? Practice. So what I did was the theoretical thing.

This is what I've done. that I just finished.

But in practice, we have actually a rule of thumb. And the rule of thumb says that I have to have, for the truncation error, for the rounding off error, I want my age to be much larger Then the var epsilon of the machine that you are working with to the power of 1 over q plus 1, where q is the accuracy order, so the order of the method that you are working with.

So in practice, how do I do?

So, can I have five minutes for an example? All over your face. So assuming that I have a tolerance of One half. times 10 to the minus six for the truncation error. Estimate when to stop decreasing edge when calculating f prime at 1/2.

for function f that is equal to sine of x using the forward difference method that we were just Looking at?

And my age is equal to 4 to the negative n.

So Maybe we, so let me stop right here. I don't think you. I think you're too tired, maybe?

So what I will do is I will restart next lecture with this example.

with a review of what I just did, and then we'll restart when this is done. Is that okay? Anything else?

No? Guys, come on.

I don't think we're going to get there.

No? No, of course not.

No, we have. But, I mean, we have. Wait, wait. So we have less than one week for the exam. Right.

Yes. Then when we And then because during Ramadan we have less than two months.

So I basically am out of three weeks of classes.

So we wouldn't do integration at all?

No, I'm thinking of integration.

Maybe-I can do something about ODEs, but for sure not PDEs. But it's, I mean, once you understand how differentiation works, which is this part, the numerical differentiation, then it's very easy to Go and, well, not very easy, but it's easier to understand the rules for integrating OEs. I'm sorry.